

DISTRICT EARLY WARNING SYSTEM

Barranquilla Convive

A digital platform for monitoring, early warning, and mental health support within the district's educational institutions.

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Project team

The project is developed by a multidisciplinary team organized under the SCRUM methodology, with clearly defined roles to ensure an agile, high-quality delivery.

Product Owner

PRODUCT

Yulianis Delgado

Defines the product vision and prioritizes the backlog based on the value delivered to end users.

Scrum Master

PROCESS

Kerlys Bello

Facilitates SCRUM ceremonies and removes obstacles to keep the team's workflow moving.

Development team

ENGINEERING

MEMBER

MAIN FOCUS

Kevin Villalobos

Platform development

Santiago Otálora

Platform development

David Carrascal

Platform development

Camilo Villalobos

Platform development

What Barranquilla Convive is about

Barranquilla Convive is a minimum viable product (MVP) designed to support the District Early Warning System for Mental Health within the city's educational institutions.

The platform centralizes the management of students, teachers, psychologists, and administrators, making it possible to identify psychosocial risk situations early through questionnaires, alerts, and follow-up processes.

Each user accesses a personalized dashboard based on their role: **teachers** monitor their students and submit alerts; **psychologists** manage intervention cases and follow-up sessions; **administrators** oversee institutions, users, and reports; and **students** complete assessments and access educational resources.

The system aims to improve communication between schools and mental health professionals, enabling timely decision-making through centralized information and statistical analysis.

03 General objective

PROJECT PURPOSE

to develop a digital platform that supports the District Early Warning System for Mental Health by centralizing student monitoring, risk alerts, psychological follow-up, and administrative management across Barranquilla's educational institutions.

04 Specific objectives

- Provide administrators with tools to manage institutions, users, permissions, reports, and system statistics.
- Enable teachers to identify students at psychosocial risk by submitting alerts and monitoring their assigned groups.
- Give psychologists a centralized environment to manage intervention cases, follow-up sessions, and recommendations.
- Allow students to complete mental health questionnaires, access educational resources, and track their own progress.
- Centralize mental health information on a secure platform that supports early intervention and data-driven decision-making.

The problem we're solving

Today, many educational institutions manage mental health referrals through manual processes: spreadsheets, emails, or paper forms, making it difficult to detect at-risk students in a timely manner. This situation creates several problems that directly affect institutions' ability to respond:

Scattered information

Lack of centralized information about students who require psychological support.

Slow communication

Delays in communication between teachers, psychologists, and administrators.

Limited follow-up

Insufficient monitoring of ongoing intervention processes.

No statistics

Difficulty generating institutional indicators and risk levels.

Without a digital platform that integrates questionnaires, alerts, follow-up, and reports, opportunities for early intervention may be missed, reducing the effectiveness of mental health support within educational institutions.

MVP modules by role

The minimum viable product defines four main roles, each with a specific set of features within the platform.

Administrator

MANAGEMENT

- Manage users and assign roles
- Approve registrations
- Manage educational institutions
- View alerts
- Generate and export reports
- Configure the system

Teacher

CLASSROOM

- Dashboard
- View assigned students
- Submit alerts
- View questionnaire results
- Access educational resources

Psychologist

INTERVENTION

- Dashboard
- View assigned cases
- Register interventions
- Schedule follow-up sessions
- Generate recommendations
- View reports

Student

SELF-ASSESSMENT

- Dashboard
- Complete psychological questionnaires
- View personal results
- Access educational resources
- View recommendations

User stories

01 As a **student**, I want to log into the platform securely, so that I can access my personal dashboard.

ACCEPTANCE CRITERIA

- ✓ The student must enter valid credentials.
- ✓ Invalid credentials must display an error message.
- ✓ Passwords must be securely encrypted.

02 As a **student**, I want to complete mental health questionnaires, so that my emotional well-being can be assessed.

ACCEPTANCE CRITERIA

- ✓ The questionnaire displays all questions.
- ✓ All required questions must be answered.
- ✓ The system stores the answers.
- ✓ The questionnaire generates a risk level.

03 As a **teacher**, I want to report an alert about a student, so that the psychology department can review the case.

ACCEPTANCE CRITERIA

- ✓ The teacher selects the student.
- ✓ The teacher enters the alert description.
- ✓ The alert is saved.
- ✓ The psychologist receives the alert.

04 As a **psychologist**, I want to review assigned alerts, so that I can begin the intervention process.

ACCEPTANCE CRITERIA

- ✓ The psychologist sees only assigned cases.
- ✓ Cases can be filtered by priority.
- ✓ The psychologist can update the case status.
- ✓ Every change is saved.

User stories — continued

05 As an **administrator**, I want to manage users, so that I can assign the correct roles.

ACCEPTANCE CRITERIA

- ✓ Create users.
- ✓ Edit users.
- ✓ Delete users.
- ✓ Assign roles.
- ✓ Approve registrations.

06 As an **administrator**, I want to generate reports, so that I can monitor overall mental health indicators.

ACCEPTANCE CRITERIA

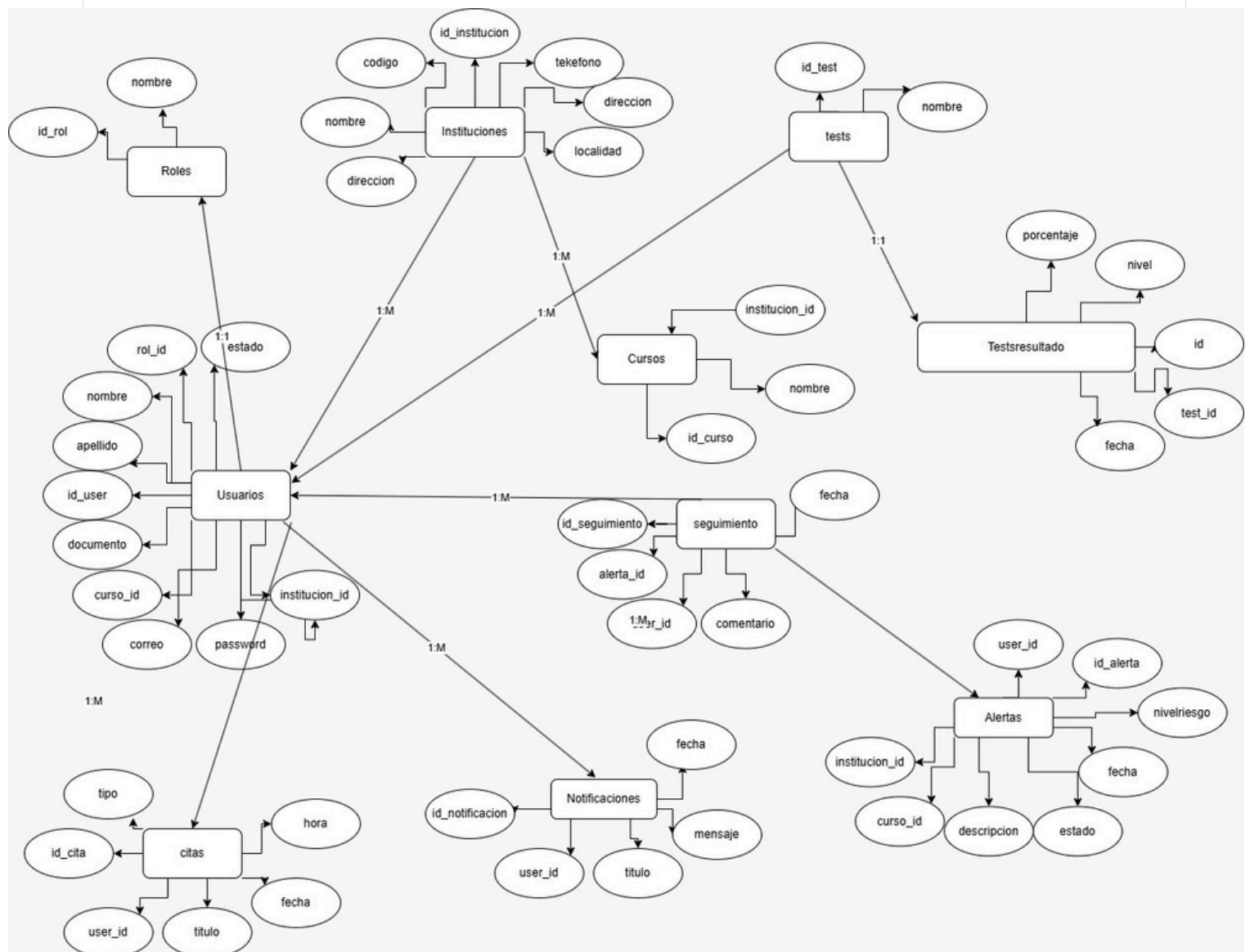
- ✓ Reports show statistics.
- ✓ Reports can be filtered.
- ✓ Reports can be exported.
- ✓ Only administrators can access reports.

07 As a **student**, I want to access educational resources, so that I can improve my emotional well-being.

ACCEPTANCE CRITERIA

- ✓ Resources are displayed by category.
- ✓ Students can open each resource.
- ✓ Access is restricted to authenticated users.

Entity-relationship model (ER)



SCRUM methodology evidence

The project was developed across four phases within each sprint cycle, following the SCRUM agile framework.

PHASE 1

Planning

- Backlog management in Jira / Trello
- Product Backlog creation
- Sprint Planning
- User story definition
- Role assignment
- Database design and Figma prototype

PHASE 2

Development

- Authentication
- Role-based dashboards
- User and institution management
- Alert management
- Reports and statistics
- Mental health questionnaires

PHASE 3

Review

- Demonstration of completed modules
- Functional testing
- User acceptance validation

PHASE 4

Retrospective

- Team feedback
- Identifying improvement opportunities
- Planning the next sprint

MVP sprint

Sprint goal

DELIVERY TARGET

Deliver the first MVP of Barranquilla Convive, including secure authentication, user management, role-based dashboards, mental health questionnaires, and early warning management.

Timeline



Resources of interest

Entity-Relationship Model

To be attached in its final high-resolution version

PENDING

GitHub repository

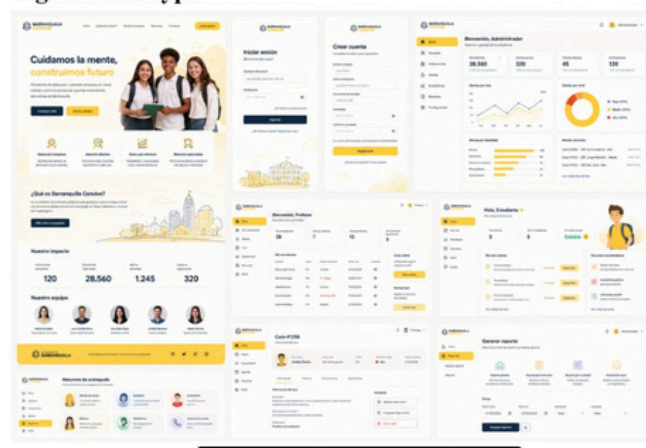
Project source code

[GITHUB.COM/ANDELGADO/
PROYECTO_INTEGRADO](https://github.com/ANDELGADO/PROYECTO_INTEGRADO)

Figma prototype

Overview of the screens designed for each platform role: public landing page, authentication, administrator dashboard, teacher dashboard, student dashboard, and case management.

Figma Prototype



Interface prototype — Barranquilla Convive, Figma